

COURSE 3 - DATA ENGINEERING 2026**C3-064B PharmaSpark Protected Review**

Open only after the matching Gate attempt/evidence is frozen.

Artifact	C3-064B-REVIEW
Status	2026 - LEARNER EDITION
Phase	C3-05 Distributed Processing with Spark 16 lessons 46-60 focused hours
Prerequisite	C3-04 Gate PASS; C2B SQL/Python/modeling foundations assumed, distributed execution taught here.
Current lane	Apache Spark / PySpark 4.2.0 Python 3.10+ Java 17+ learner local[2]
Brand	Charcoal #1F2937 Teal #14B8A6 Soft Blue #3B82F6 AMOLED #000

House teaching rule: mental picture -> tiny worked example -> guided real Spark workflow -> expected result -> why it worked -> break/fix -> independent changed task -> validation -> physical-plan/production reasoning -> evidence. A green Spark job is not mastery if grain, quality, plan or reproducibility is wrong.

What a passing submission proves

- Correct business grain and source/reject reconciliation before any performance discussion.
- Dimension key is unique before join; no DISTINCT used to hide fan-out.
- Grouped and window outputs have explicit, independently validated grain.
- Plan evidence connects Exchange/join/operator choices to code/data.
- Hot-key diagnosis distinguishes data quality from legitimate skew.
- Performance change has a hypothesis, one controlled variable and correctness parity.
- Tests protect rows/schema/grain/rejection/totals or equivalent case-specific invariants.
- Job is parameterized and reproducible from a clean environment.
- No secrets/private paths/fabricated runtime screenshots are included.

Routing

PASS only when every required competency passes and zero critical failures remain. PARTIAL/FAIL -> targeted remediation + changed micro-retest. A fresh full Gate B/C opens only by explicit assignment; failed attempts remain append-only.